

Green Central Banking

Vitor M. Dias

Abstract

In 2017, the Network of Central Banks and Supervisors for Greening the Financial System (NGFS) was established to promote sustainability in financial regulation. This article provides the first empirical investigation into whether NGFS member countries have reduced their greenhouse gas (GHG) emissions. Using panel data from 138 countries over 21 years, two-way fixed-effect models reveal that NGFS members, on average, emit significantly less GHG than non-members, controlling for ecological, economic, and political factors. These findings support the literature on the positive impact of international networks on policy outcomes. By proposing transition pathways and fostering policy alignment through financial frameworks that integrate climate risks, the NGFS has promoted financial and social-ecological sustainability deliberations among its members.. However, as the NGFS is predominantly composed of financial regulators from the Global North, the results also highlight the unequal financial capacities and policy implementation between the Global North and South. Therefore, this study discusses the potentials of such international networks in the need for more inclusive global governance mechanisms.

Introduction

Data and Methods

EDGAR emissions data come from here: https://edgar.jrc.ec.europa.eu/dataset_ghg80.

World Bank data come from: <https://databank.worldbank.org/source/world-development-indicators>.

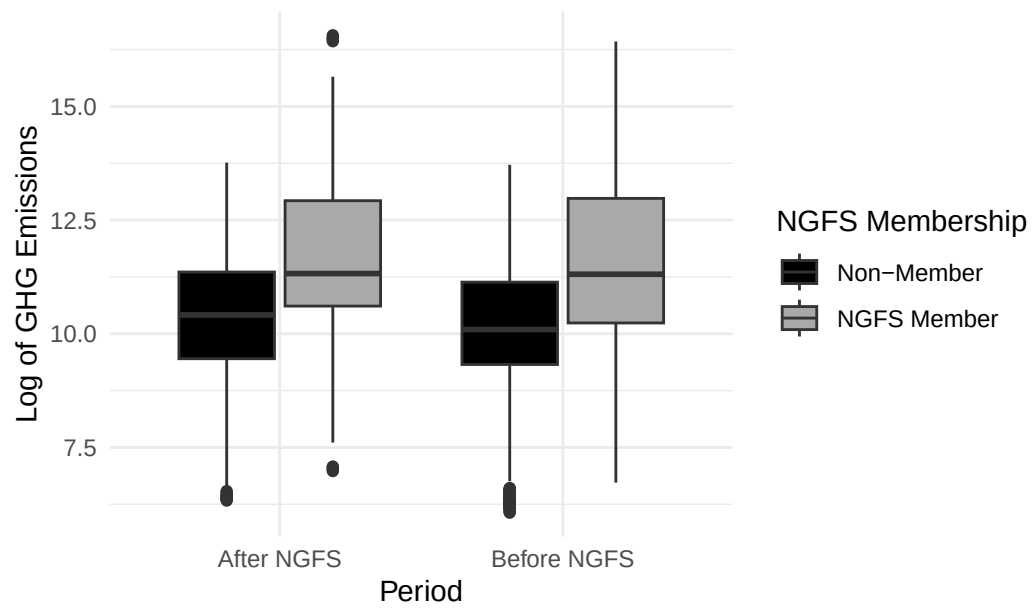
V-Dem data come from: <https://www.v-dem.net/data/the-v-dem-dataset/>.

Analysis

Table 1: Descriptive Statistics of Country x Years by Network for Greening the Financial System Membership between 2001 and 2021 (N = 2,856)

Variable	Member		Non-Member		Means Diff.	SE
	Mean/N	SD/%	Mean/N	SD/%		
Greenhouse gas emissions	11.80	1.83	10.80	1.85	-1.01***	0.14
Trade dependence	4.41	0.59	4.29	0.50	-0.12**	0.04
GDP per capita	9.87	1.06	8.49	1.45	-1.38***	0.08
GDP per capita growth	3.93	0.11	3.95	0.12	0.02*	0.01
Natural resources	-0.90	2.48	0.60	2.17	1.50***	0.18
Population density	4.51	1.41	4.12	1.38	-0.39***	0.10
Agricultural land	5.81	2.00	6.11	1.85	0.30*	0.15
Forest land	5.52	1.83	5.02	2.27	-0.50***	0.14
Corruption index (0-1)	0.24	0.25	0.50	0.31	0.26***	0.02
Deliberative democracy index (0-1)	0.60	0.24	0.44	0.25	-0.15***	0.02
Administrative capacity index (0-1)	0.65	0.20	0.52	0.20	-0.13***	0.02
East and South Asia and Pacific	40	20.5	401	15.1		
Europe and Central Asia	97	49.7	848	31.9		
Latin America and Caribbean	23	11.8	355	13.3		
Middle East and Africa	29	14.9	1021	38.4		
North America	6	3.1	36	1.4		

GHG Emissions Before and After NGFS Creation



Survival analysis predicting the probability of becoming an NGFS member

Probability of Joining the Network for Greening the Financial System by Country-Level

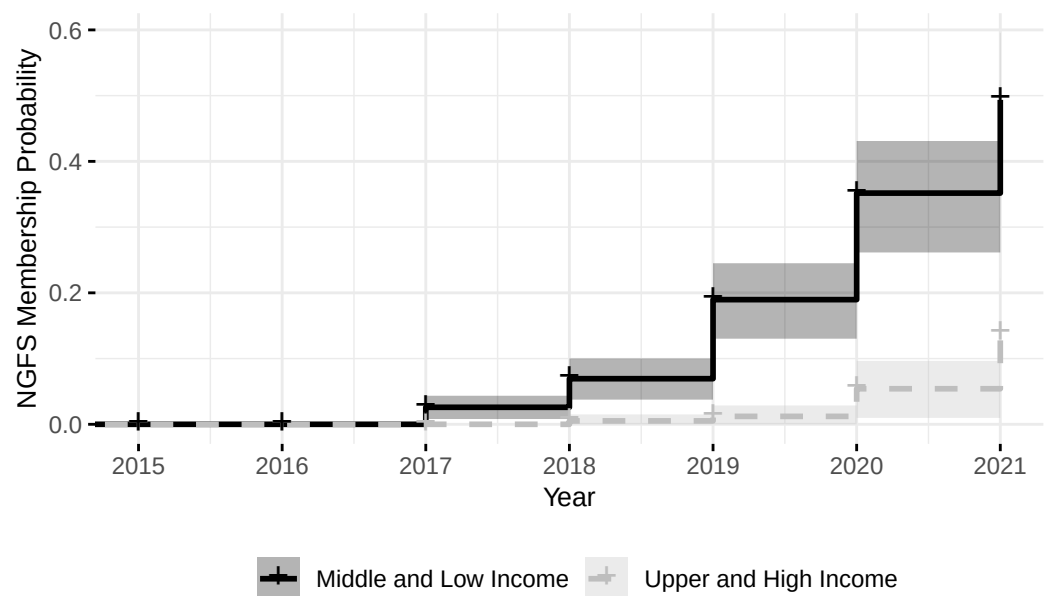
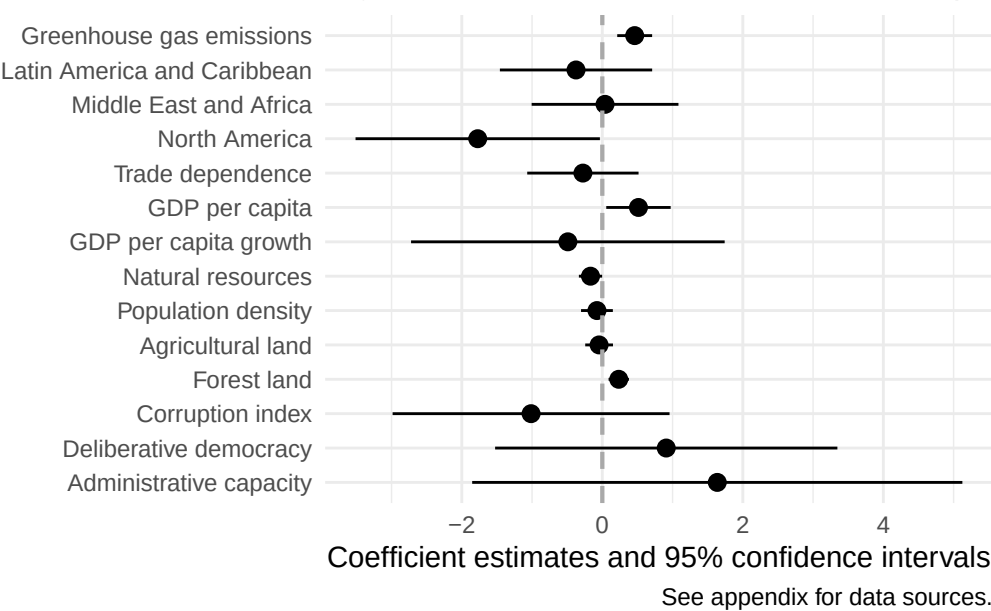


Figure 1: Predictors of NGFS Membership



Panel data analysis predicting greenhouse gas emissions based on NGFS membership

Robustness checks

	Baseline	Ecological and Economic Indicators	Political Indicators
Greenhouse gas emissions	0.28*** (0.08)	0.31** (0.11)	0.46*** (0.13)
Latin America and Caribbean	-0.05 (0.44)	-0.68 (0.51)	-0.37 (0.55)
Middle East and Africa	-0.95* (0.42)	-0.36 (0.50)	0.04 (0.53)
North America	0.97 (0.80)	-1.77* (0.89)	-1.77* (0.89)
Trade dependence		-0.30 (0.39)	-0.28 (0.40)
GDP per capita		1.06*** (0.18)	0.51* (0.23)
GDP per capita growth		-0.44 (1.04)	-0.49 (1.14)
Natural resources		-0.17* (0.07)	-0.17* (0.08)
Population density		-0.07 (0.11)	-0.08 (0.12)
Agricultural land		0.01 (0.09)	-0.05 (0.10)
Forest land		0.29*** (0.07)	0.23** (0.07)
Corruption index			-1.01 (1.01)
Deliberative democracy			0.91 (1.24)
Administrative capacity			1.64 (1.78)
Num.Obs.	2731	2731	2731
BIC	748.3	710.0	721.3

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Source: The data have been collected from the World Bank, the Network for Greening the Financial System, and the Varieties of Democracy (V-Dem) Project. Europe is the reference group for the region variable.